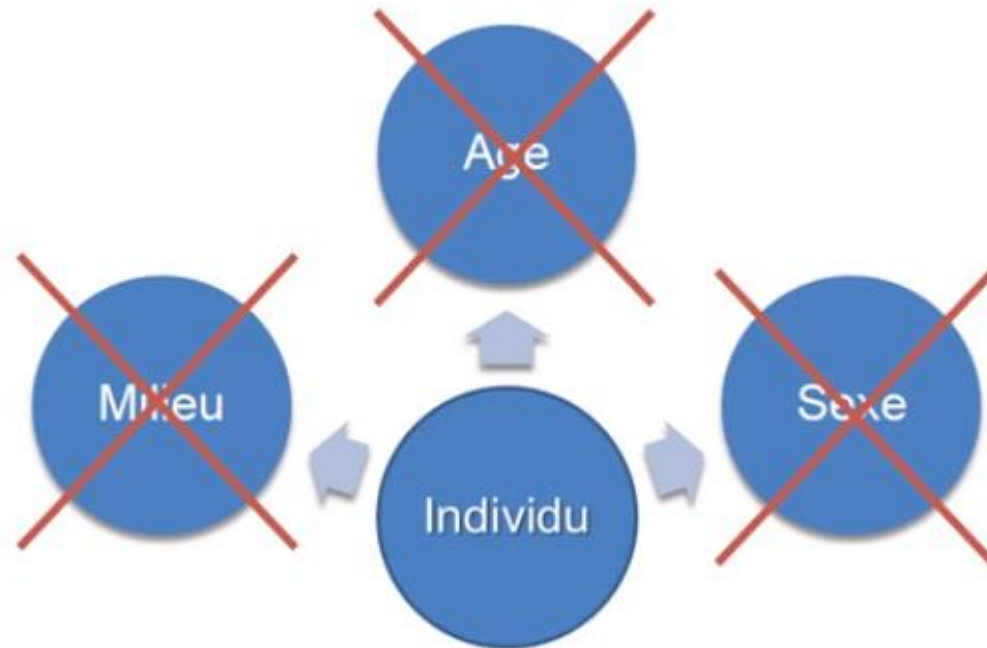


SOCIAL NETWORK ANALYSIS

INTRODUCTION

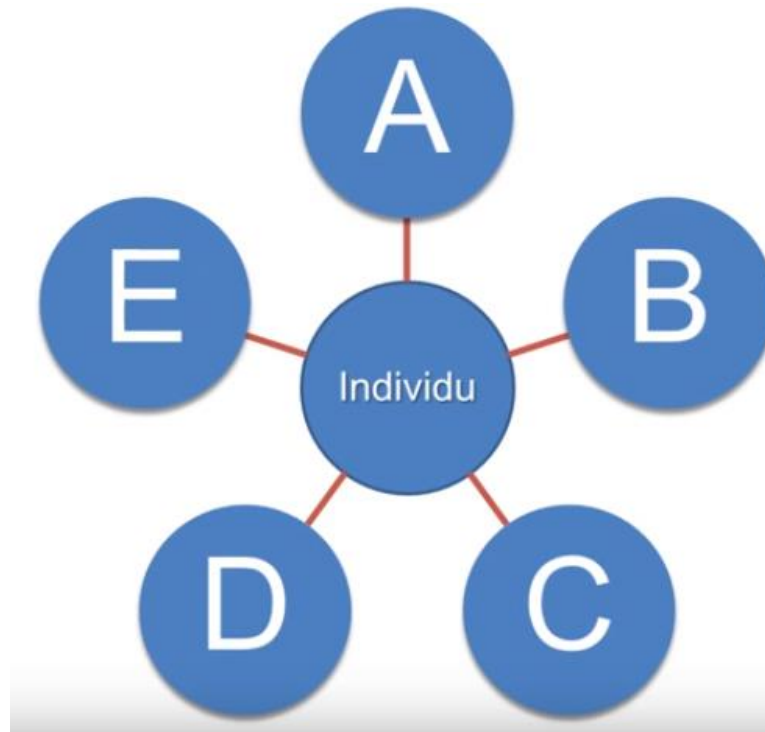
- What is a social network ?



Individual attributes are not interest in this context

INTRODUCTION

- What is a social network ?



**The most interesting thing is its relationship
with other individuals**

INTRODUCTION

■ Social network emergence

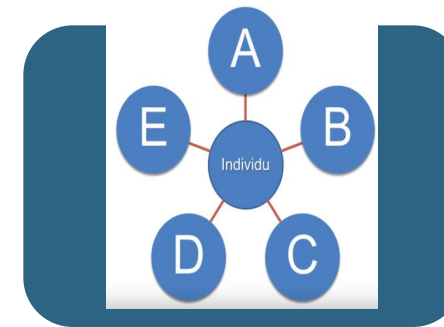
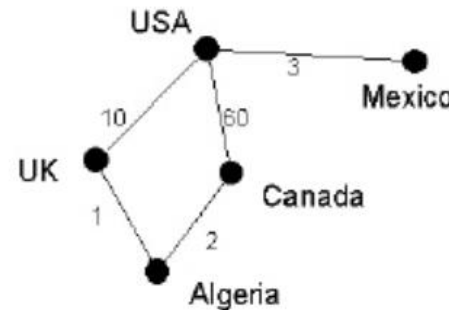
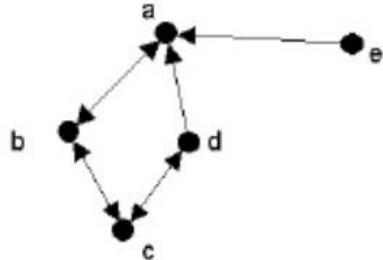


- Involvement of social connections in many phenomena:
 - Diffusion problem : rumor, disease, ...etc.
 - Influence phenomenon: social connection can determine a behavior.

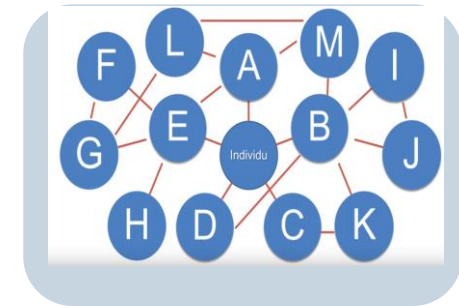
INTRODUCTION

■ Social network objective

- Studying dependencies between given individual actions and the structure (nature) of its relationship with other individuals.
- Different types of relationships:
 - Direct / Indirect relationship (Information flow)
 - Weighted / Unweighted relationships (interactions frequency)
 - Unique / Multiple relationships



Direct
relationship



Indirect
relationship

INTRODUCTION

■ Social network objective

Different types of entities (actors):

- One type of entity to describe a network of friends, or a network of blogs or web pages
- Two types of entities to describe mediated relationships: authors - articles, actors - movies

INTRODUCTION

- Social network: Definition

Social network : set of social actors who have direct or indirect relationships with each other.

Studying social network = Studying digital social network ??



INTRODUCTION

- Social network: Definition

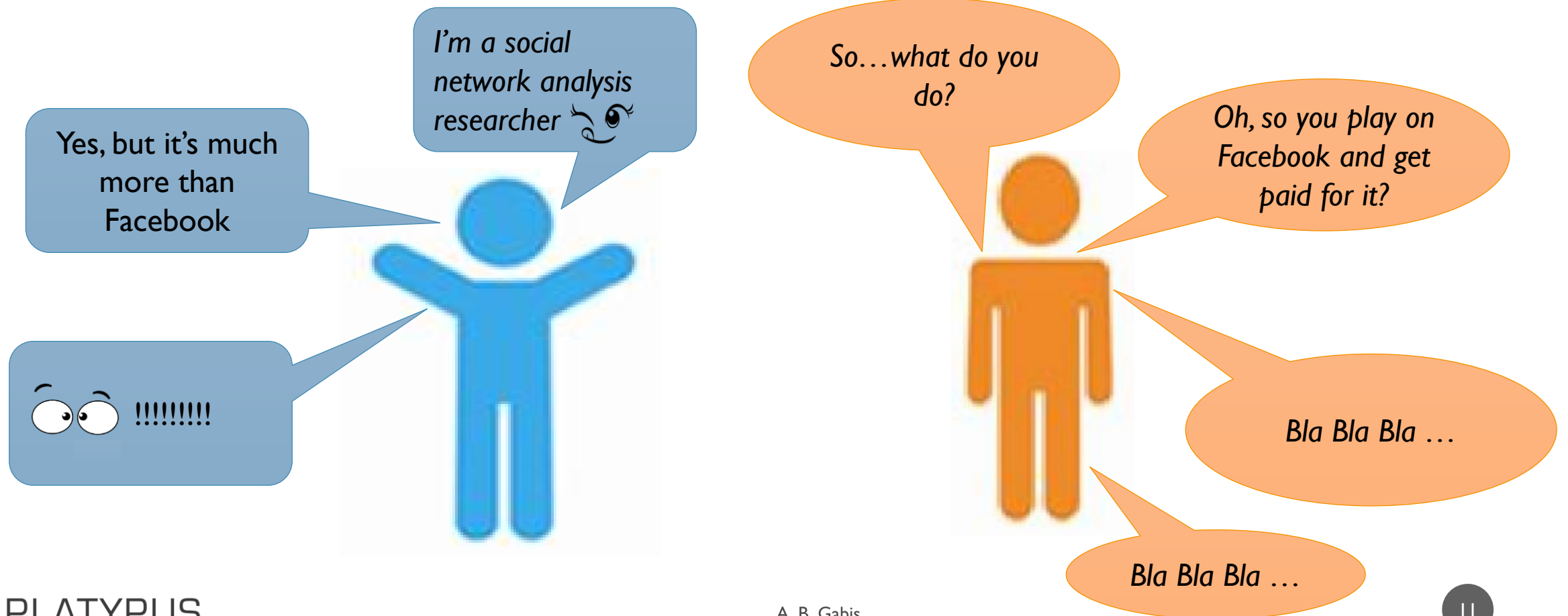
Study of human relationships by means of **graph theory**



SOCIAL NETWORK ANALYSIS (SNA)

SOCIAL NETWORK ANALYSIS (SNA)

■ What is a social network analysis ?



SOCIAL NETWORK ANALYSIS (SNA)

■ History

- The analysis of social networks is not a new discipline:
 - 1933: sociogram of Jacob Levy Moreno to objectify the **dynamics of the group**
 - 1954: John A. Barnes introduces **the term "social network"**
 - 1967: Stanley Milgram: effect of the "little world", **six degrees** of separation
 - 1973: Mark L. Granovetter: the **strength of weak contacts**
 - 1998: Google's PageRank: measurement of **relevance** in the **scheduling of responses to queries**

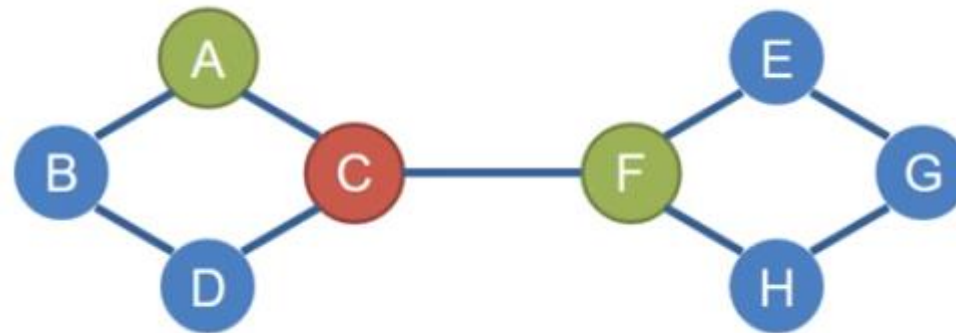
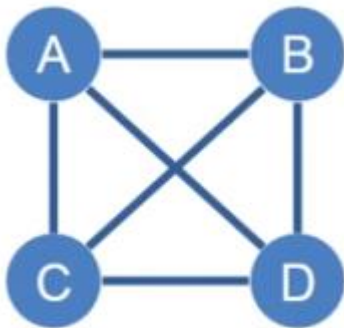


SOCIAL NETWORK ANALYSIS (SNA)

■ Social network structure

➤ $SN = G(V, E)$ with:

- V = a set of vertex : individuals
- E = a set of edges : relationship between individuals (also called ties)



SOCIAL NETWORK ANALYSIS (SNA)

■ Research questions

Three big questions are usually asked in social network studies:

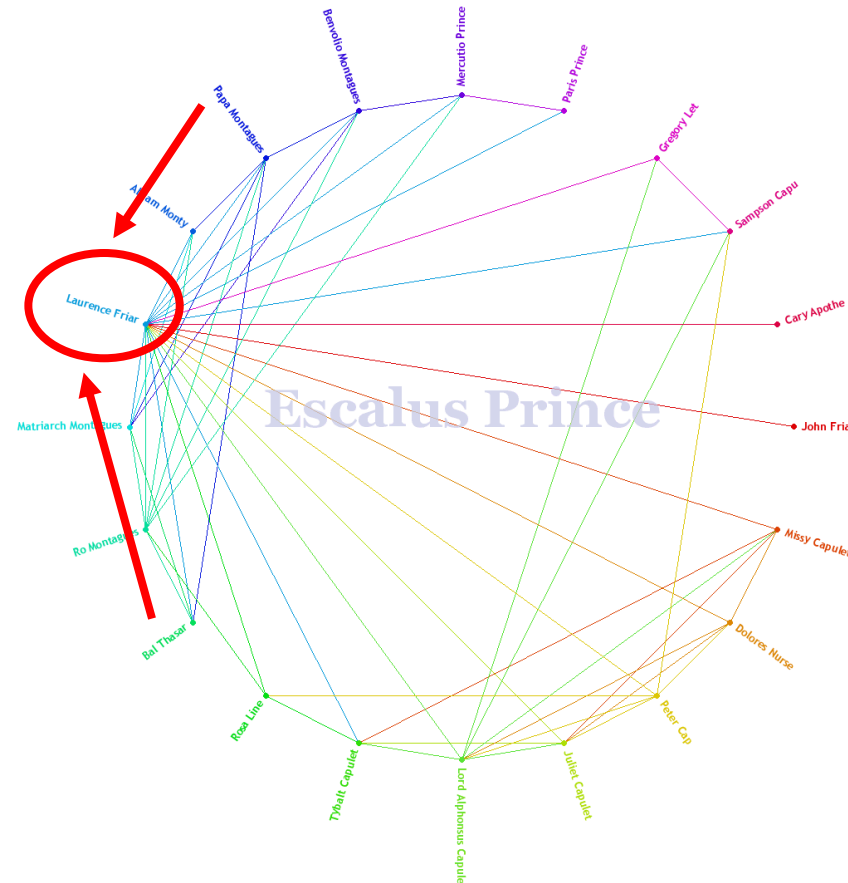
- The question of **homophilia**: Do my relationships tend to resemble me and according to which criteria: gender, age, level of education, etc. ?
- The question of **centrality**: Which are the most influential individuals, the best placed in a given network?
- A question closely linked to the previous one, the **circulation of information**: is the network effective in disseminating information, a rumor, an innovation?



SOCIAL NETWORK ANALYSIS (SNA)

■ What's the use of social networks ?

- Facebook friend's wheel



Source

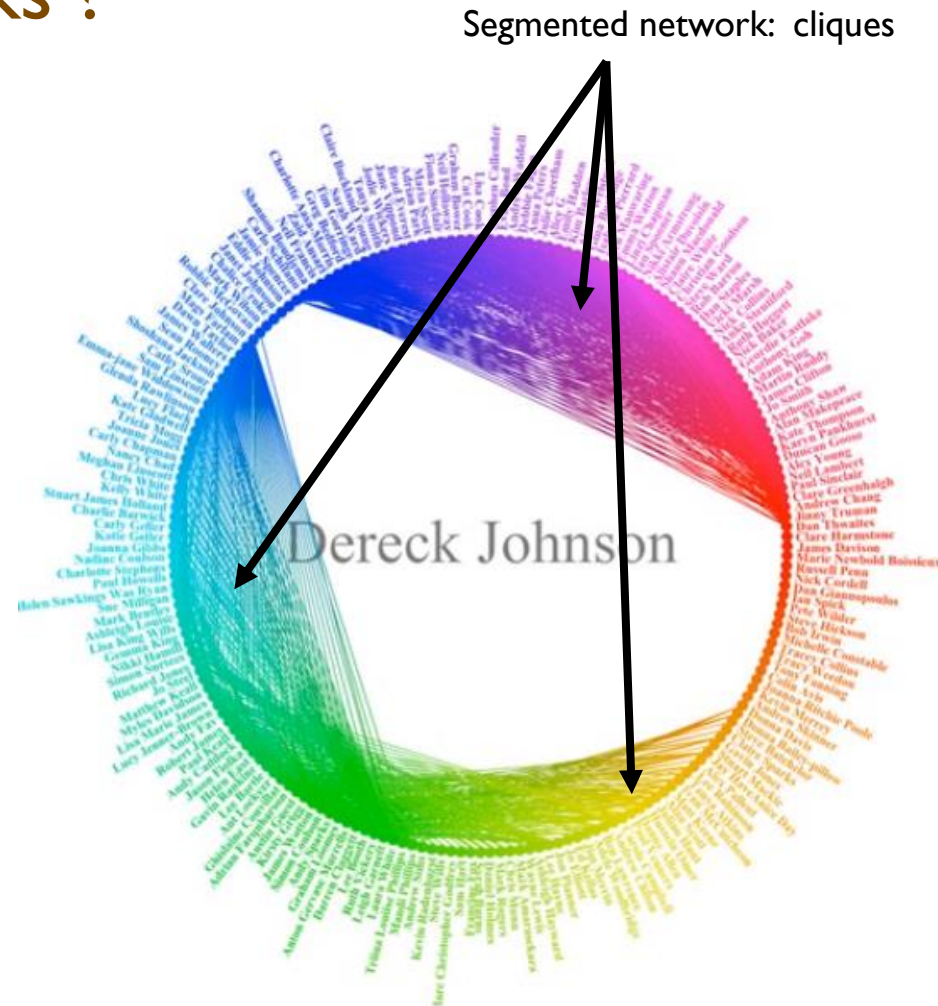
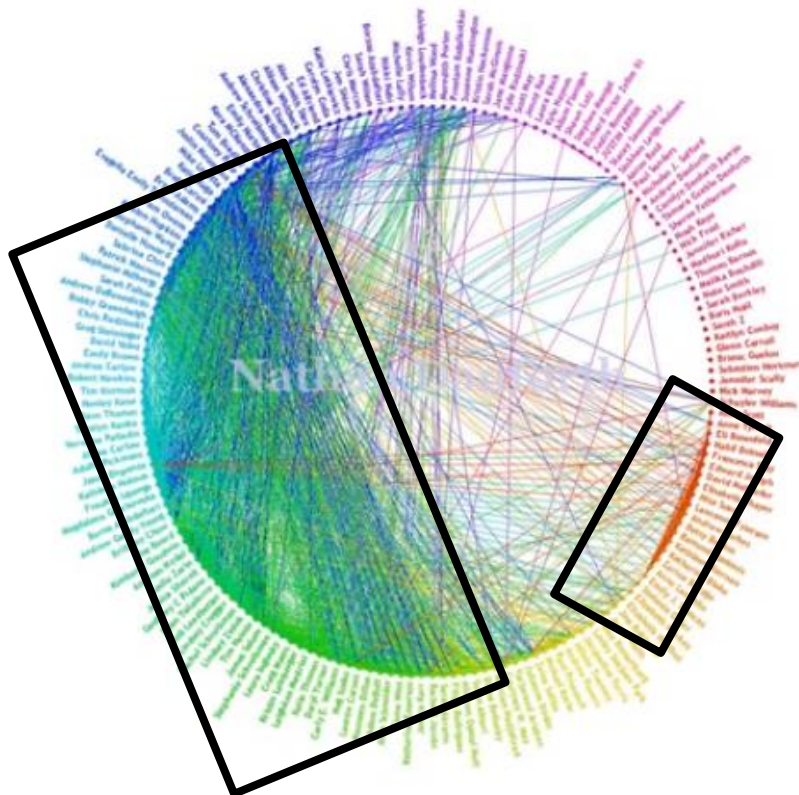


PLATYPUS

SOCIAL NETWORK ANALYSIS (SNA)

■ What's the use of social networks ?

- Facebook friend's wheel

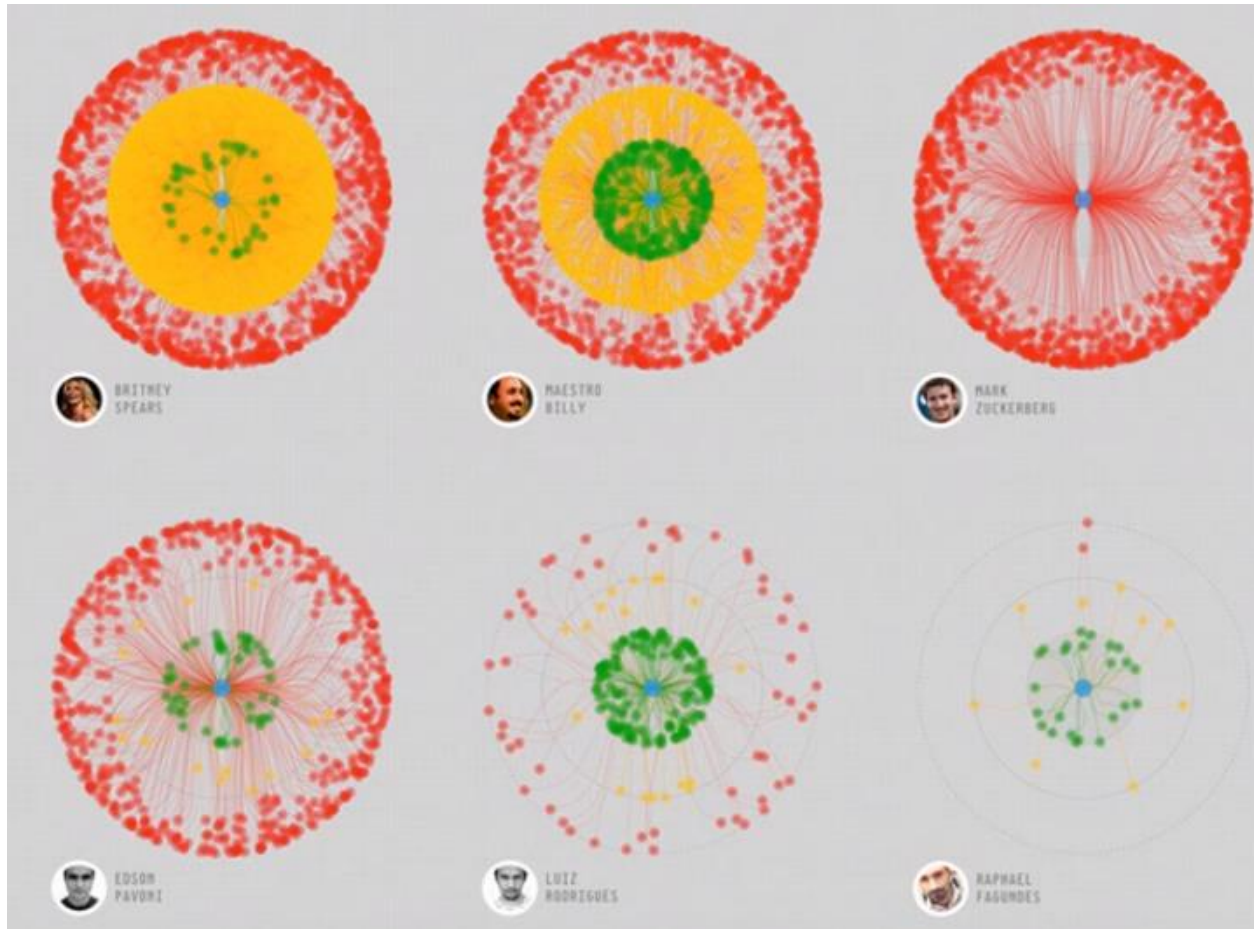


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SOCIAL NETWORK ANALYSIS (SNA)

■ What's the use of social networks ?

- Google+ circle of trust



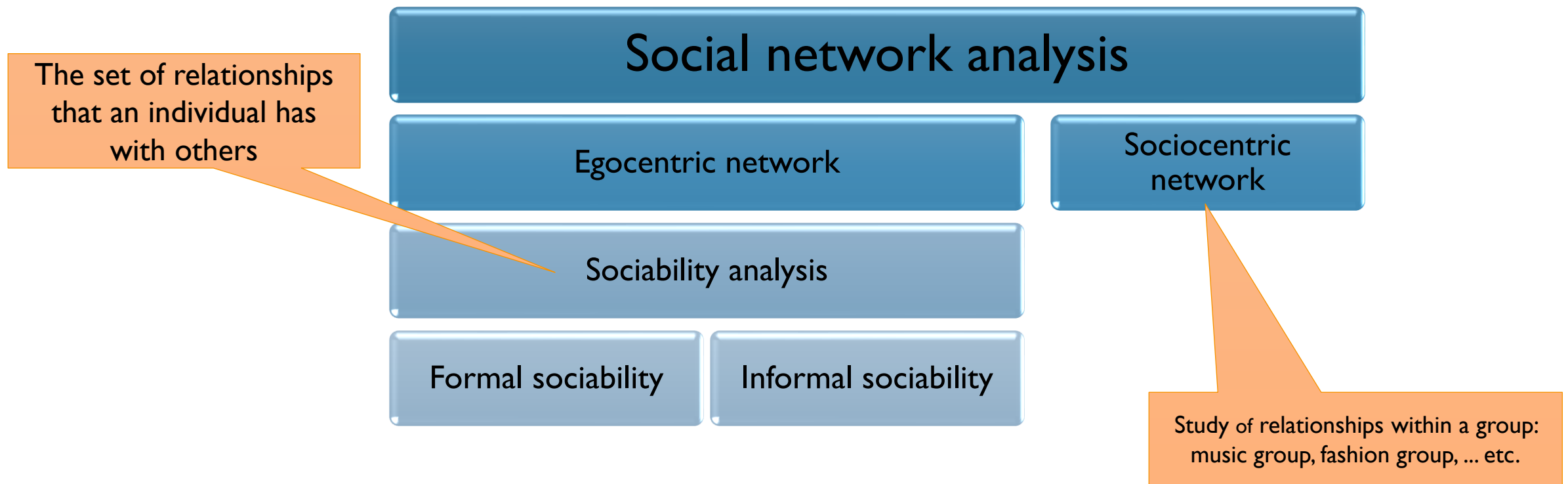
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PLATYPUS

SOCIAL NETWORK ANALYSIS (SNA)

■ Social networks analysis types



SOCIAL NETWORK ANALYSIS (SNA)

■ Social networks analysis types

➤ Sociocentric network

- Sociocentric = Whole networks creates one network
- “If your research question is about different patterns of interaction within defined groups”
- Network boundaries often clear: Everyone working at a particular company , Everyone in one classroom of a school; ...

➤ Egocentric network

- Egocentric = Personal networks
- Creates many stand alone networks
- “If your research question is about phenomena of or affecting individual entities across different settings”
- Network boundaries: You have decisions to make : people who spoke with an individual in the day



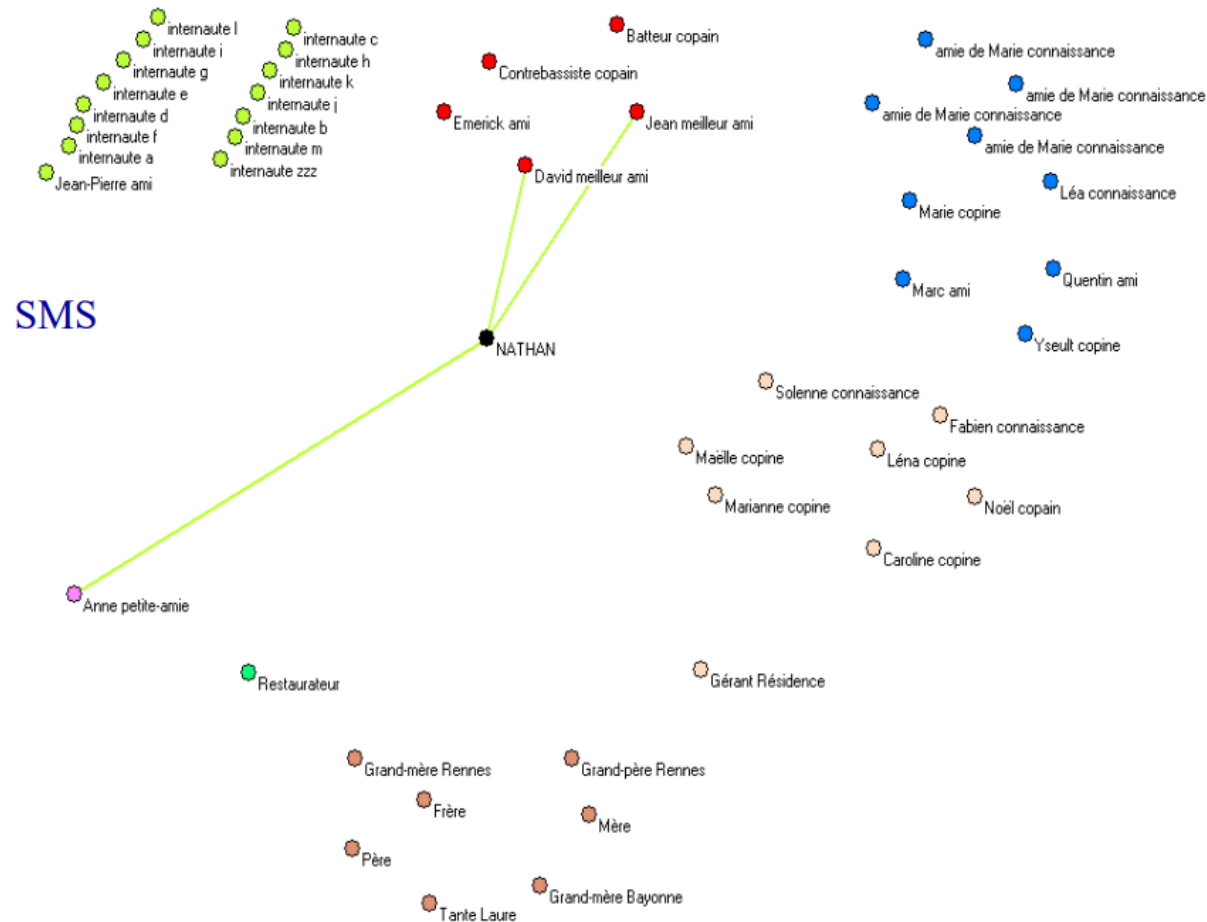
SOCIAL NETWORK ANALYSIS (SNA)

■ Egocentric network

- Also called: personal network
- One node called **ego** is at center, surrounded by other nodes called **alter**
- By "surrounded" we mean all alters surrounding the ego have links to the ego.
- We use "node" and "link" instead of "vertex" and "edge" when talking about graph

SOCIAL NETWORK ANALYSIS (SNA)

■ Egocentric network



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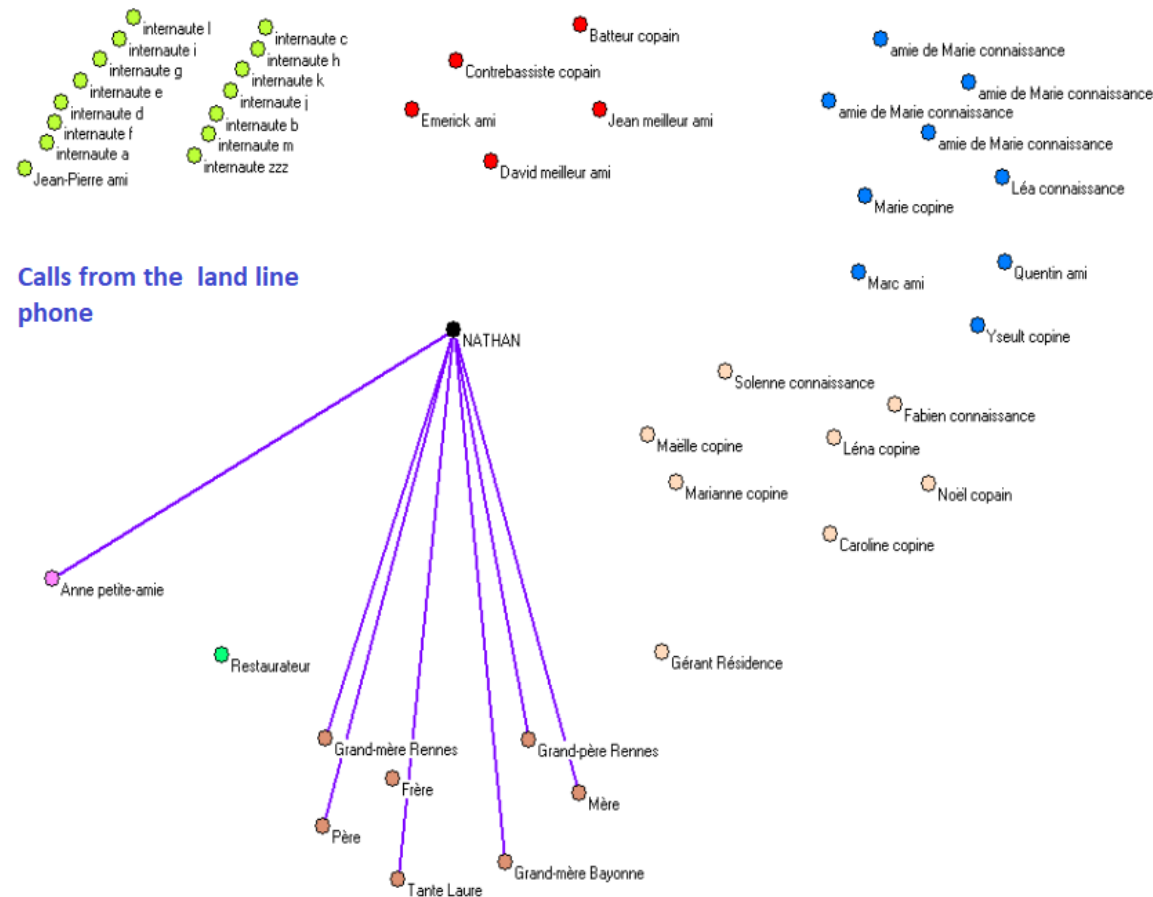


PLATYPUS

A. B. Gabis

SOCIAL NETWORK ANALYSIS (SNA)

■ Egocentric network

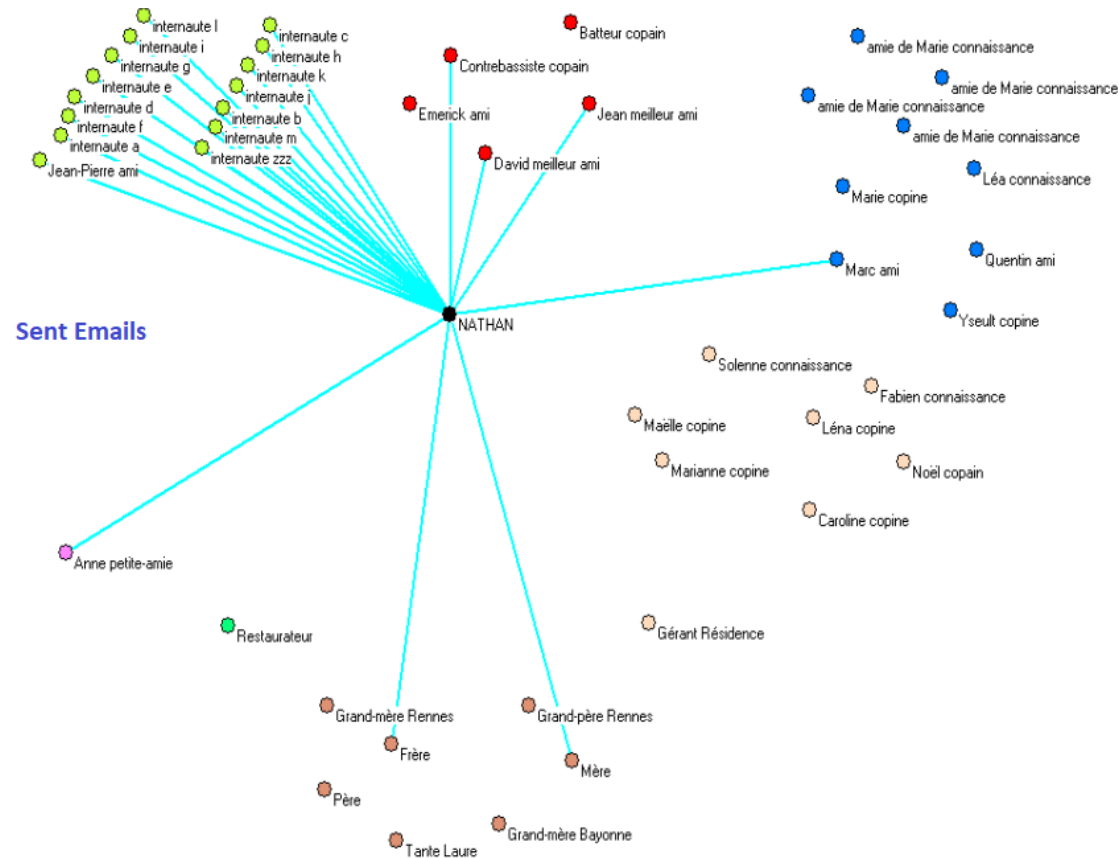


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SOCIAL NETWORK ANALYSIS (SNA)

■ Egocentric network



Source



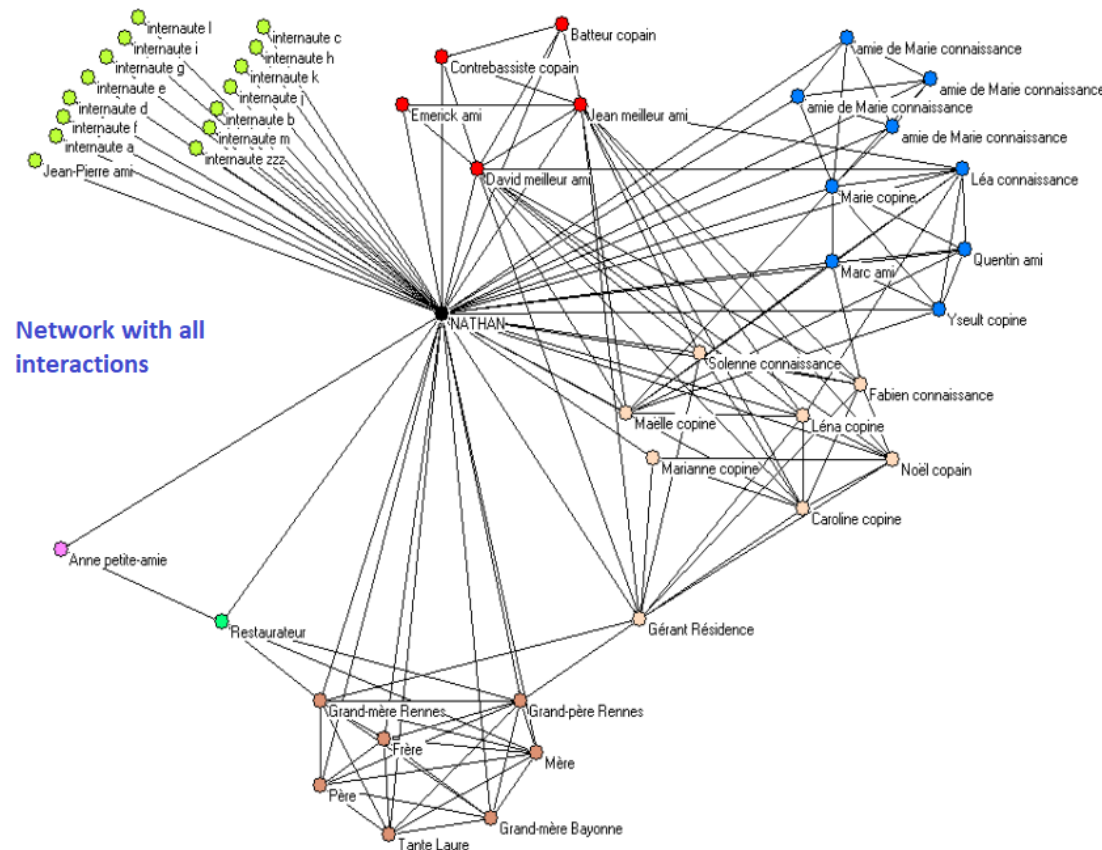
PLATYPUS

A. B. Gabis

27

SOCIAL NETWORK ANALYSIS (SNA)

■ Egocentric network



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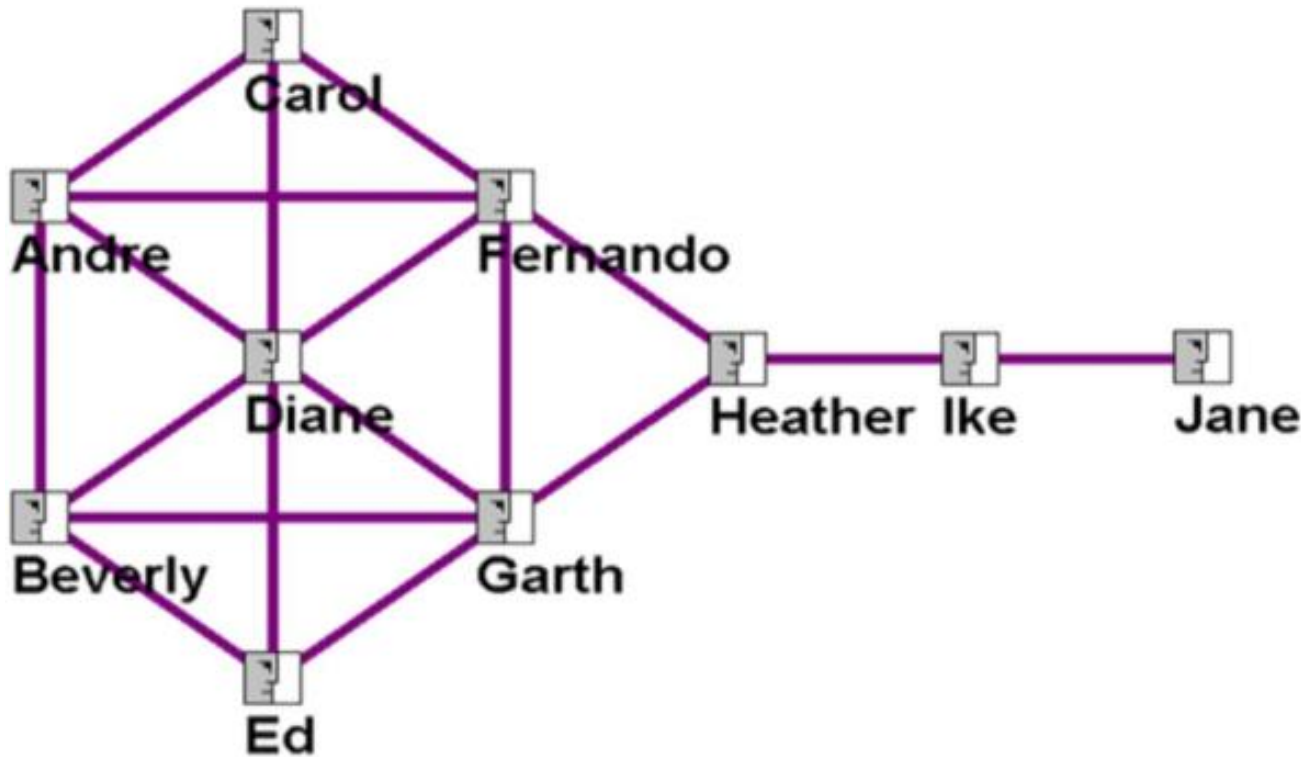


- Egocentric network



SOCIAL NETWORK ANALYSIS (SNA)

■ Sociocentric (global) network

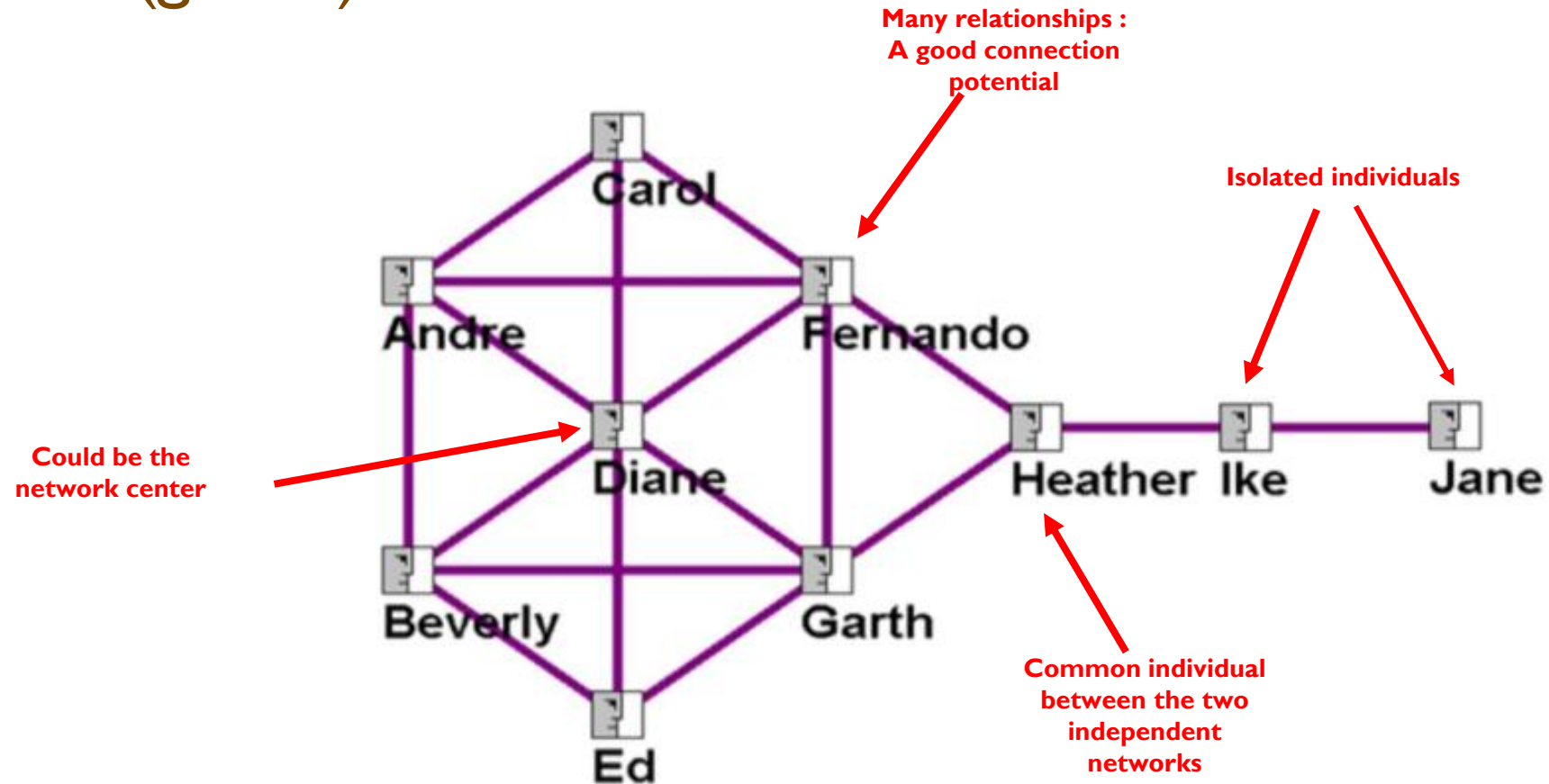


- Who is the "hub" in this network?
- Who controls access to information?
- Who knows best what happens?
- Who are the peripheral actors? Are all equally important?

Source

SOCIAL NETWORK ANALYSIS (SNA)

■ Sociocentric (global) network



Source

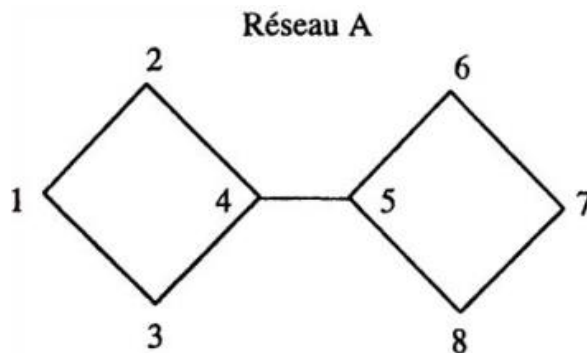


STRUCTURAL ANALYSIS

STRUCTURAL ANALYSIS

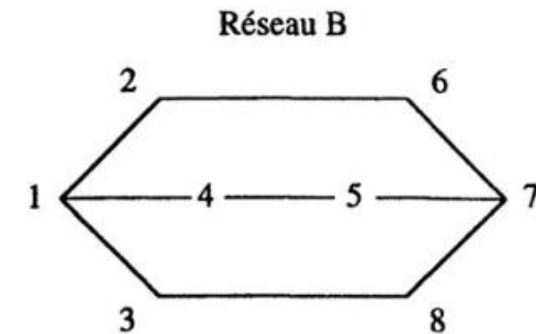
■ Social networks: Global structure

- **Size:** It defines the number of edges (ties).
- **Degree** (of a node): Defines the number of edges that connect to it.
- **Distance:** The distance between two vertices (nodes) is the number of edges of the shortest path to go from one to the other. (also called **geodesic distance**)



A	B
4,57	3,86

Average distance

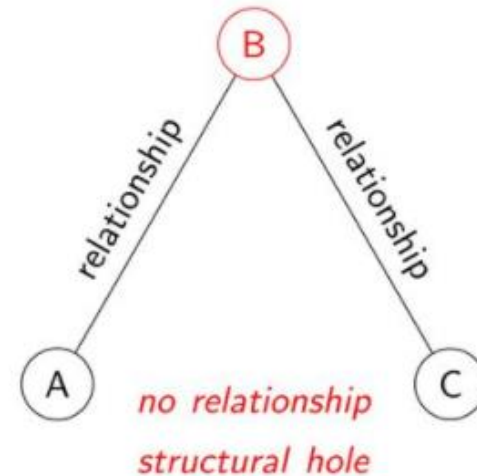


- **Diameter:** To find the diameter of a graph, first find the shortest path between each pair of vertices. The greatest length of any of these paths is the diameter of the graph.

STRUCTURAL ANALYSIS

■ Social networks: Global structure

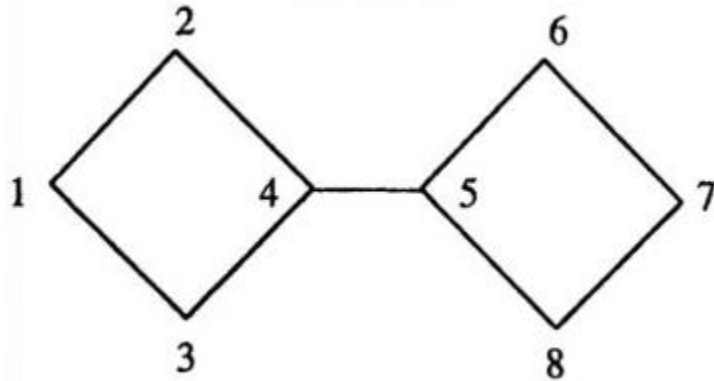
- **Structural holes:** refers to the empty space between two relationships in a group. It is an absence of relationship. This lack of relationship between two individuals allows a third individual to be an intermediary and therefore take advantage of the situation.
- Its advantages are of three kinds:
 - ✓ Faster access to information;
 - ✓ Better information;
 - ✓ Controlling the dissemination of information



STRUCTURAL ANALYSIS

■ Social networks: Global structure

- **Density:** it is the ratio between the number of relationships observed and the number of possible relationships in a network. This allows to define if the number of links is too many or not enough compared to the needs.



$$\delta = \frac{k}{\frac{n(n-1)}{2}} = \frac{9}{28} = 0.32$$

The closer this number is to 1, the denser is the network and then the fewer are structural holes. As a result, individual reputation will be known to everyone as information flows easily.

STRUCTURAL ANALYSIS

■ Social networks: Global structure

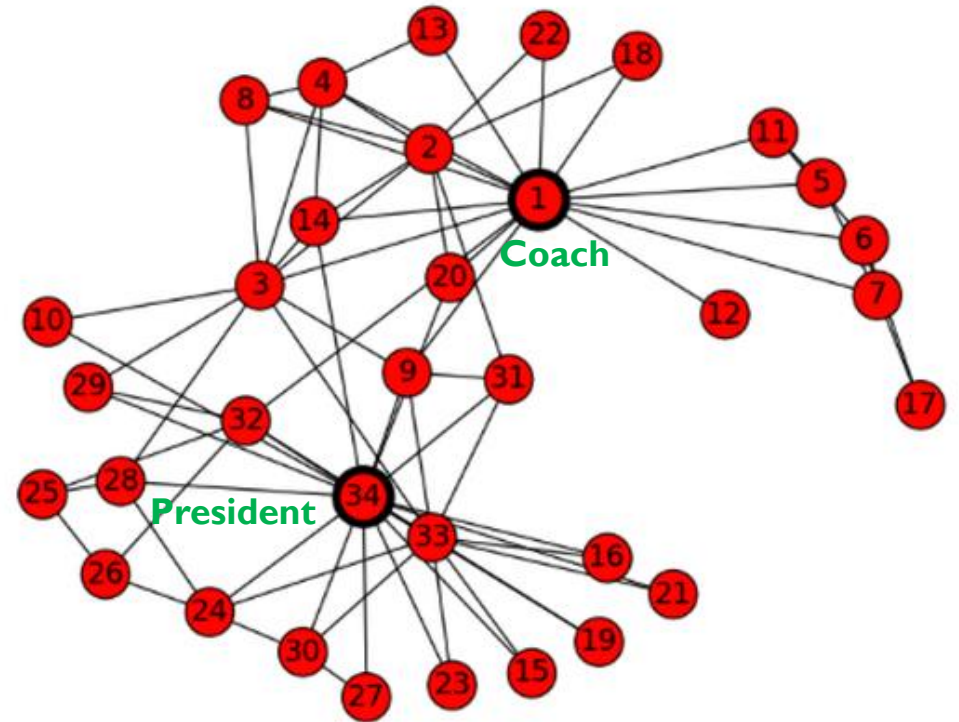
- **Centrality**: it aims to identify influential actors in a network. As a general rule, in a network, the people through whom information travels most often are the most influential. So, we are talking about central people.
- **Different definitions of centrality:**
 - ✓ Degree
 - ✓ Closeness
 - ✓ PageRank
 - ✓ Betweenness

STRUCTURAL ANALYSIS

■ Social networks: Global structure

➤ Centrality: Degree

- It is the measure of the simplest and most intuitive centrality. An individual can be considered important if this individual has a lot of direct contact.
- The degree corresponds to the sum of the direct links of a node.
- This measure can be directional, taking into account the number of incoming links or outgoing links to the node (in-degree, out-degree).



STRUCTURAL ANALYSIS

■ Social networks: Global structure

➤ Centrality: Closeness

- An individual will have an important centrality in the sense of closeness, if he is close to all other individuals, either by direct links or by indirect links.
- Closeness is the inverse of the sum of the direct and indirect links of a node to all other nodes.

Closeness du noeud i : $\frac{1}{\sum_{j=1}^n d_{ij}}$ où d_{ij} est le plus court chemin de i à j .

STRUCTURAL ANALYSIS

■ Social networks: Global structure

➤ Centrality: PageRank/EigenVector

- This measurement is based on Google's Pagerank algorithm, which is behind the success of its recommendation engine to determine relevant sites when searching the net.
- It is closer to the centrality degree measure, because it also takes into account the number of direct links of a node.
- **BUT**, it assumes that the links have different weights. A link with a node will have more weight if this node itself has a lot of direct links, or a low weight if it is rather on the periphery of the network.
- An individual is important, in the sense of Pagerank, if he knows individuals who are important.

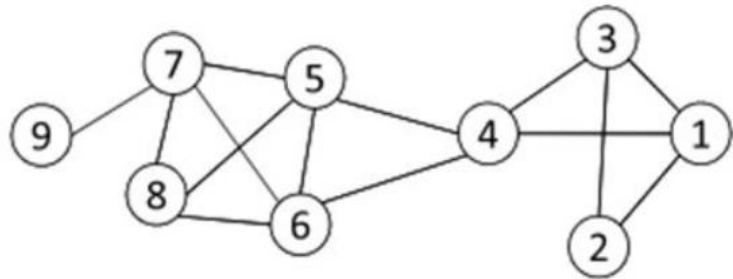


STRUCTURAL ANALYSIS

■ Social networks: Global structure

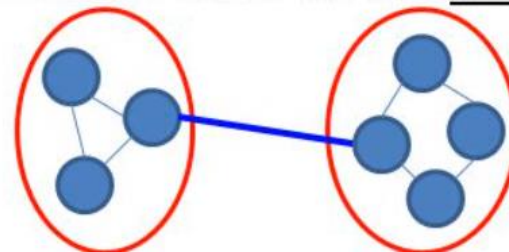
➤ Centrality: Betweenness edge

- The strength of a tie can be measured by **edge betweenness**
- **Edge betweenness**: the number of shortest paths that pass along with the edge



The edge betweenness of $e(1, 2)$ is 4 ($=6/2 + 1$), as all the shortest paths from 2 to $\{4, 5, 6, 7, 8, 9\}$ have to either pass $e(1, 2)$ or $e(2, 3)$, and $e(1, 2)$ is the shortest path between 1 and 2

- The edge with higher betweenness tends to be the bridge between two communities.



STRUCTURAL ANALYSIS

■ Social networks: Global structure

➤ The strength of weak ties

Two major types of ties in a social network:

- **Strong ties:** frequent meetings, in-depth exchanges (Personal relationships)
- **Weak ties:** brief, casual contacts, support, contact (Professional relationships)

To find a job, the weak links are more important,

- They are more numerous
- They form "local bridges" and create shorter paths
- While networks with strong links are highly integrated, information is not leave the group

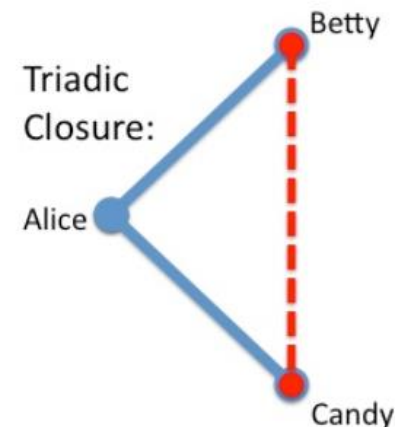


STRUCTURAL ANALYSIS

■ Social networks: Global structure

➤ Triadic closure

- A characteristic feature of social networks is the property of *Triadic Closure*
- *Informally: Two people who have a common friend are likely to become friends themselves. The more closer they are to their common friend, the more likely is it that they become friends themselves*



Source



STRUCTURAL ANALYSIS

■ Social networks: Global structure

➤ **Dunbar Number**

- The number of Dunbar is the maximum number of individuals with whom a person can simultaneously maintain a stable human relationship.
- In social network, the maximum number of strong ties an individual can maintain is usually between 10 and 20



STRUCTURAL ANALYSIS

■ Social networks: Global structure

➤ Six degrees of separation

- If we take two random people from the American population they are connected by a chain of six relationships (1967)
- In 2011 a study on Facebook shows that now there would be a degree of separation of 4.74 (Lars Backstrom, Paolo Boldi, Marco Rosa, Johan Ugander, Sebastiano Vigna).

Source



CLUSTERING AND COMMUNITIES ANALYSIS

CLUSTERING AND COMMUNITIES ANALYSIS

■ Main idea

- We do not look for who belongs to a family, ethnic community ...
- Highlight an emerging category, based on topology analysis
 - (In practice) Looking for dense areas in the network.

CLUSTERING AND COMMUNITIES ANALYSIS

■ Communities detection

- **Community:** It is formed by individuals such that those within a group interact with each other more frequently than with those outside the group a.k.a. group, cluster, cohesive subgroup, module in different contexts.
- **Community detection:** discovering groups in a network where individuals' group memberships are not explicitly given.
- **Communities types:** 2 main types
 - **Explicit groups:** Formed by user subscriptions
 - **Implicit groups:** Implicitly formed by social interactions.



CLUSTERING AND COMMUNITIES ANALYSIS

■ Communities detection

- Some social networks (ex: social media sites) allow people to join groups, is it necessary to extract groups based on network topology?
 - Not all sites provide community platform
 - Not all people want to make effort to join groups
 - Groups can change dynamically
- Network interaction provides rich information about the relationship between users
 - Can complement other kinds of information, e.g. user profile
 - Help network visualization and navigation
 - Provide basic information for other tasks, e.g. recommendation



CLUSTERING AND COMMUNITIES ANALYSIS

■ Communities detection

➤ More generally:

- behavior analysis
- location-based interaction analysis
- recommender systems development
- link prediction
- customer interaction and analysis & marketing
- media use
- Security
- Social studies



CLUSTERING AND COMMUNITIES ANALYSIS

■ Taxonomy community criteria

Criteria vary depending on the tasks. Community detection methods can be divided into 4 categories:

Community detection methods

Node-centric
community

Group-
centric
commu
nity

Network-centric community

Hierarchy-
centric
community

Complete
Mutuality

Finding the
Maximum
Clique

Clique
Percolation
Method
(CPM)

Density-
based
group

Clustering
based
method

Latent
space
models

Block
model
approxima
tion

Spectral
clustering

Modularity
maximizati
on

Divisive
Hierarchical
Clustering

Agglomerativ
e
Hierarchical
Clustering



CLUSTERING AND COMMUNITIES ANALYSIS

■ Taxonomy community criteria

➤ Node-centric community detection

Each node in groups satisfies certain properties:

- Complete mutuality
 - ✓ Cliques
- Reachability of members
 - ✓ K-clique, k-clan, k-club
- Nodal degrees
 - ✓ K-plex, k-core

Commonly used in traditional social network analysis

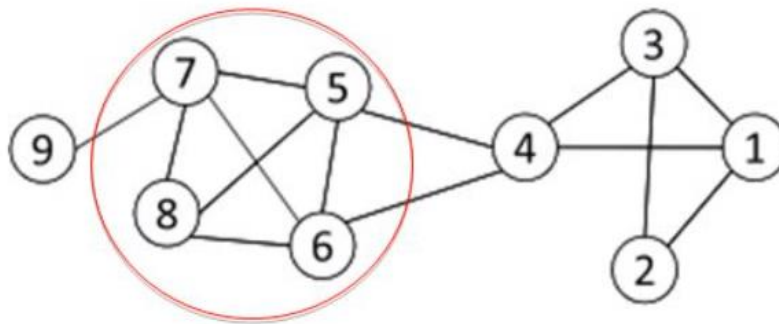


CLUSTERING AND COMMUNITIES ANALYSIS

■ Taxonomy community criteria

➤ Node-centric community detection: **Complete mutuality**

- **Clique**: a maximum complete subgraph in which all nodes are adjacent to each other



Nodes 5, 6, 7 and 8 form a clique

- NP-hard to find the maximum clique in a network
- Straightforward implementation to find cliques is very expensive in time complexity

CLUSTERING AND COMMUNITIES ANALYSIS

■ Taxonomy community criteria

➤ Node-centric community detection: Finding the maximum clique

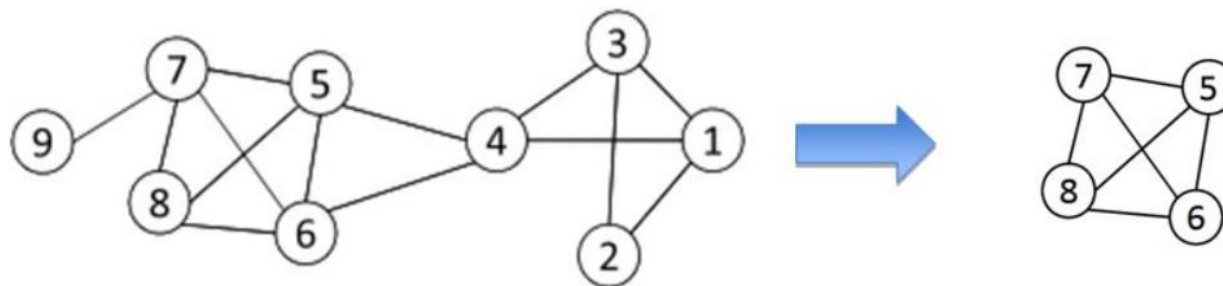
- In a clique of size k , each node maintains degree $\geq k-1$
 - Nodes with degree $< k-1$ will not be included in the maximum clique
- Recursively apply the following pruning procedure
 - Sample a sub-network from the given network, and find a clique in the sub-network, say, by a greedy approach
 - Suppose the clique above is size k , in order to find out a *larger* clique, all nodes with degree $\leq k-1$ should be removed.
- Repeat until the network is small enough
- Many nodes will be pruned as social media networks follow a power law distribution for node degrees



CLUSTERING AND COMMUNITIES ANALYSIS

■ Taxonomy community criteria

➤ Node-centric community detection: **Maximum clique example**



- Suppose we sample a sub-network with nodes {1-9} and find a clique {1, 2, 3} of size 3
- In order to find a clique >3 , remove all nodes with degree $\leq 3-1=2$
 - Remove nodes 2 and 9
 - Remove nodes 1 and 3
 - Remove node 4

CLUSTERING AND COMMUNITIES ANALYSIS

■ Taxonomy community criteria

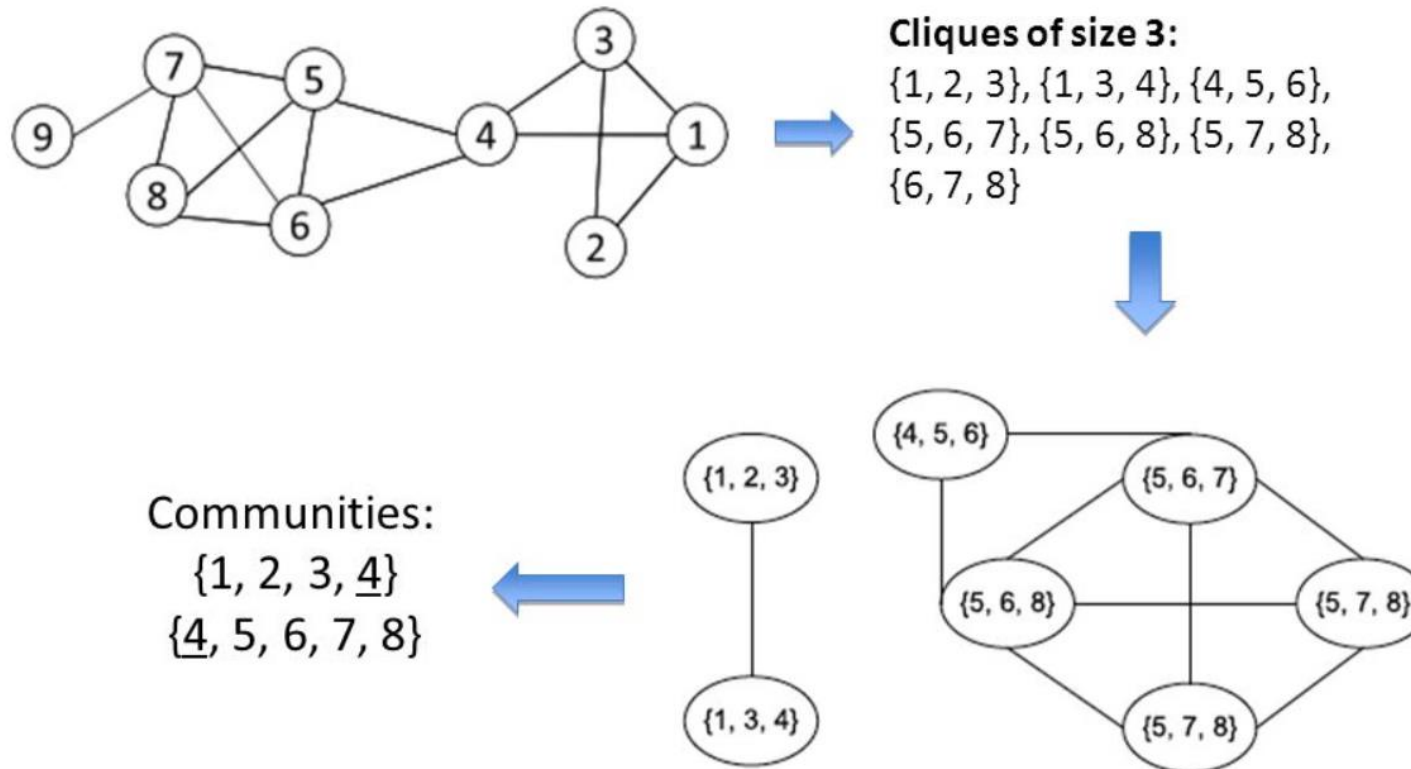
➤ Node-centric community detection: **Clique Percolation Method (CPM)**

- CPM is such a method to find **overlapping** communities
 - **Input**
 - A parameter k , and a network
 - **Procedure**
 - Find out all cliques of size k in a given network
 - Construct a clique graph. Two cliques are adjacent if they share $k-1$ nodes
 - Each connected components in the clique graph form a community

CLUSTERING AND COMMUNITIES ANALYSIS

■ Taxonomy community criteria

➤ Node-centric community detection: **CPM** example

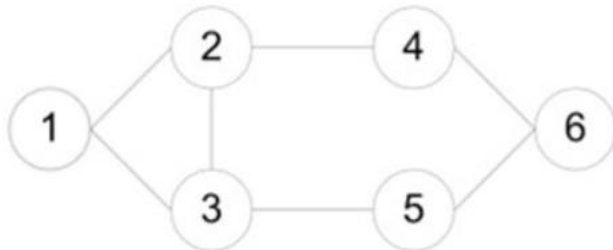


CLUSTERING AND COMMUNITIES ANALYSIS

■ Taxonomy community criteria

➤ Node-centric community detection: **Reachability (k-clique, k-club)**

- Any node in a group should be reachable in k hops
- **k-clique**: a maximal subgraph in which the largest geodesic distance between any two nodes $\leq k$
- **k-club**: a substructure of diameter $\leq k$



Cliques: $\{1, 2, 3\}$

2-cliques: $\{1, 2, 3, 4, 5\}, \{2, 3, 4, 5, 6\}$

2-clubs: $\{1, 2, 3, 4\}, \{1, 2, 3, 5\}, \{2, 3, 4, 5, 6\}$

- A k -clique might have diameter larger than k in the subgraph
 - E.g. $\{1, 2, 3, 4, 5\}$
- Commonly used in traditional SNA

CLUSTERING AND COMMUNITIES ANALYSIS

- **Taxonomy community criteria**

- **Group-centric community detection:**

Consider the connections within a group as a whole. The group has to satisfy certain properties without zooming into node-level.

E.g. the group density $\geq$ a given threshold

CLUSTERING AND COMMUNITIES ANALYSIS

■ Taxonomy community criteria

➤ Group-centric community detection: **Density-based group**

- A subgraph $G_s(V_s, E_s)$ is a γ - *dense* **quasi-clique** if

$$\frac{2|E_s|}{|V_s|(|V_s| - 1)} \geq \gamma$$

where the denominator is the maximum number of degrees.

- A similar strategy to that of cliques can be used
 - Sample a subgraph, and find a maximal γ - *dense* quasi-clique (say, of size $|V_s|$)
 - Remove nodes with degree less than the average degree

$$< |V_s|\gamma \leq \frac{2|E_s|}{|V_s|-1}$$



CLUSTERING AND COMMUNITIES ANALYSIS

■ Taxonomy community criteria

➤ Network-centric community detection

- Network-centric criterion needs to consider the connections within a network globally
- Goal: partition nodes of a network into disjoint sets
- Approaches:
 - (1) Clustering based on vertex similarity
 - **(2) Latent space models (multi-dimensional scaling)**
 - (3) Block model approximation
 - **(4) Spectral clustering**
 - **(5) Modularity maximization**



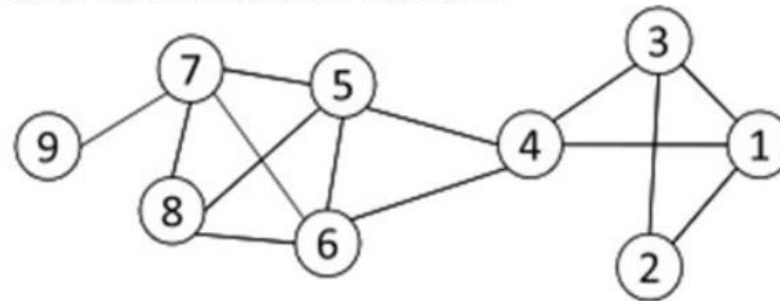
CLUSTERING AND COMMUNITIES ANALYSIS

■ Taxonomy community criteria

➤ Network-centric community detection: **Clustering-based on Vertex Similarity**

- Apply k-means or similarity-based clustering to nodes
- Vertex similarity is defined in terms of **the similarity of their neighborhood**
- **Structural equivalence**: two nodes are structurally equivalent iff they are connecting to the same set of actors

Nodes 1 and 3 are
structurally equivalent;
So are nodes 5 and 6.



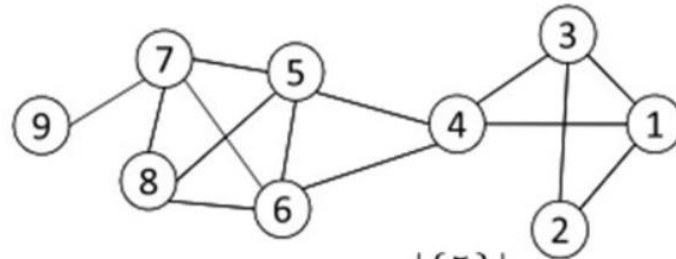
- Structural equivalence is too restrict for practical use.

CLUSTERING AND COMMUNITIES ANALYSIS

■ Taxonomy community criteria

➤ Network-centric community detection: **Vertex Similarity**

- Jaccard Similarity $Jaccard(v_i, v_j) = \frac{|N_i \cap N_j|}{|N_i \cup N_j|}$
- Cosine similarity $Cosine(v_i, v_j) = \frac{|N_i \cap N_j|}{\sqrt{|N_i| \cdot |N_j|}}$



$$Jaccard(4, 6) = \frac{|\{5\}|}{|\{1, 3, 4, 5, 6, 7, 8\}|} = \frac{1}{7}$$

$$cosine(4, 6) = \frac{1}{\sqrt{4 \cdot 4}} = \frac{1}{4}$$

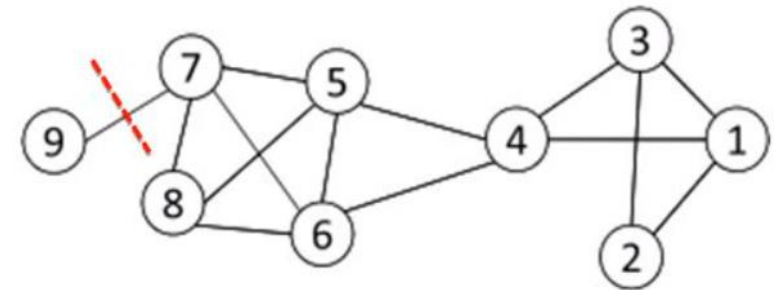
CLUSTERING AND COMMUNITIES ANALYSIS

■ Taxonomy community criteria

➤ Network-centric community detection: **Spectral clustering**

- **Cut**

- Most interactions are within group whereas interactions between groups are few
- community detection → **minimum cut problem**
- **Cut**: A partition of vertices of a graph into two disjoint sets
- **Minimum cut problem**: find a graph partition such that the number of edges between the two sets is minimized



CLUSTERING AND COMMUNITIES ANALYSIS

■ Taxonomy community criteria

➤ Network-centric community detection: **Spectral clustering**

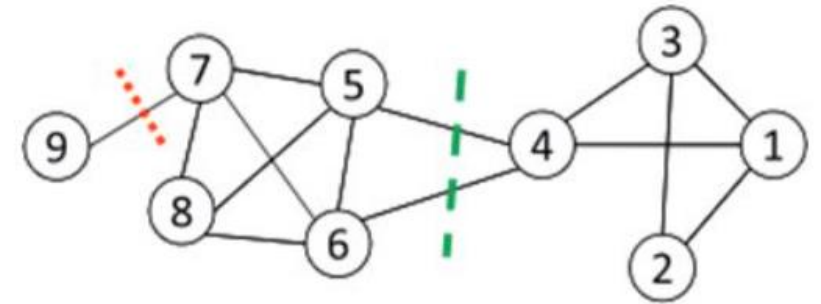
- **Ratio cut & Normalized cut**

- **Minimum cut often** returns an imbalanced partition, with one set being a singleton, e.g. node 9
- Change the objective function to consider community size

$$\text{Ratio Cut}(\pi) = \frac{1}{k} \sum_{i=1}^k \frac{\text{cut}(C_i, \bar{C}_i)}{|C_i|},$$

$$\text{Normalized Cut}(\pi) = \frac{1}{k} \sum_{i=1}^k \frac{\text{cut}(C_i, \bar{C}_i)}{\text{vol}(C_i)}$$

C_i : a community
 $|C_i|$: number of nodes in C_i
 $\text{vol}(C_i)$: sum of degrees in C_i



CLUSTERING AND COMMUNITIES ANALYSIS

■ Taxonomy community criteria

➤ Network-centric community detection: **Spectral clustering**

• **Ratio cut & Normalized cut example**

For partition in red: π_1

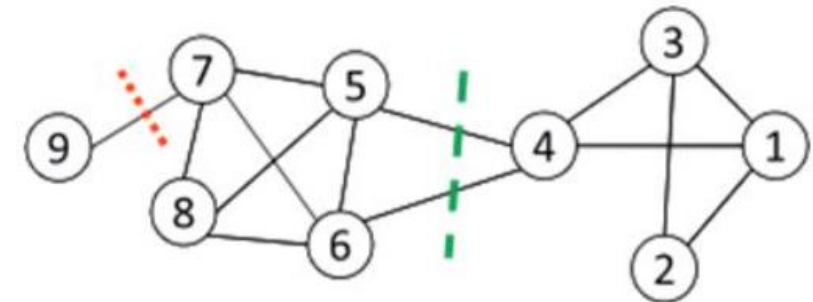
$$\text{Ratio Cut}(\pi_1) = \frac{1}{2} \left(\frac{1}{1} + \frac{1}{8} \right) = 9/16 = 0.56$$

$$\text{Normalized Cut}(\pi_1) = \frac{1}{2} \left(\frac{1}{1} + \frac{1}{27} \right) = 14/27 = 0.52$$

For partition in green: π_2

$$\text{Ratio Cut}(\pi_2) = \frac{1}{2} \left(\frac{2}{4} + \frac{2}{5} \right) = 9/20 = 0.45 < \text{Ratio Cut}(\pi_1)$$

$$\text{Normalized Cut}(\pi_2) = \frac{1}{2} \left(\frac{2}{12} + \frac{2}{16} \right) = 7/48 = 0.15 < \text{Normalized Cut}(\pi_1)$$



Both ratio cut and normalized cut prefer a balanced partition

CLUSTERING AND COMMUNITIES ANALYSIS

■ Taxonomy community criteria

➤ Network-centric community detection: **Spectral clustering**

- Both ratio cut and normalized cut can be reformulated as

$$\min_{S \in \{0,1\}^{n \times k}} \text{Tr}(S^T \tilde{L} S)$$

- Where $\tilde{L} = \begin{cases} D - A & \text{graph Laplacian for ratio cut} \\ I - D^{-1/2} A D^{-1/2} & \text{normalized graph Laplacian} \end{cases}$
 $D = \text{diag}(d_1, d_2, \dots, d_n)$ A diagonal matrix of degrees

- **Spectral relaxation:** $\min_S \text{Tr}(S^T \tilde{L} S) \quad s.t. \quad S^T S = I_k$
- Optimal solution: top eigenvectors with the smallest eigenvalues

CLUSTERING AND COMMUNITIES ANALYSIS

■ Taxonomy community criteria

➤ Network-centric community detection: **Spectral clustering**

- **More simply**

(See more details [here](#) and [there](#))

To perform a spectral clustering we need 3 main steps:

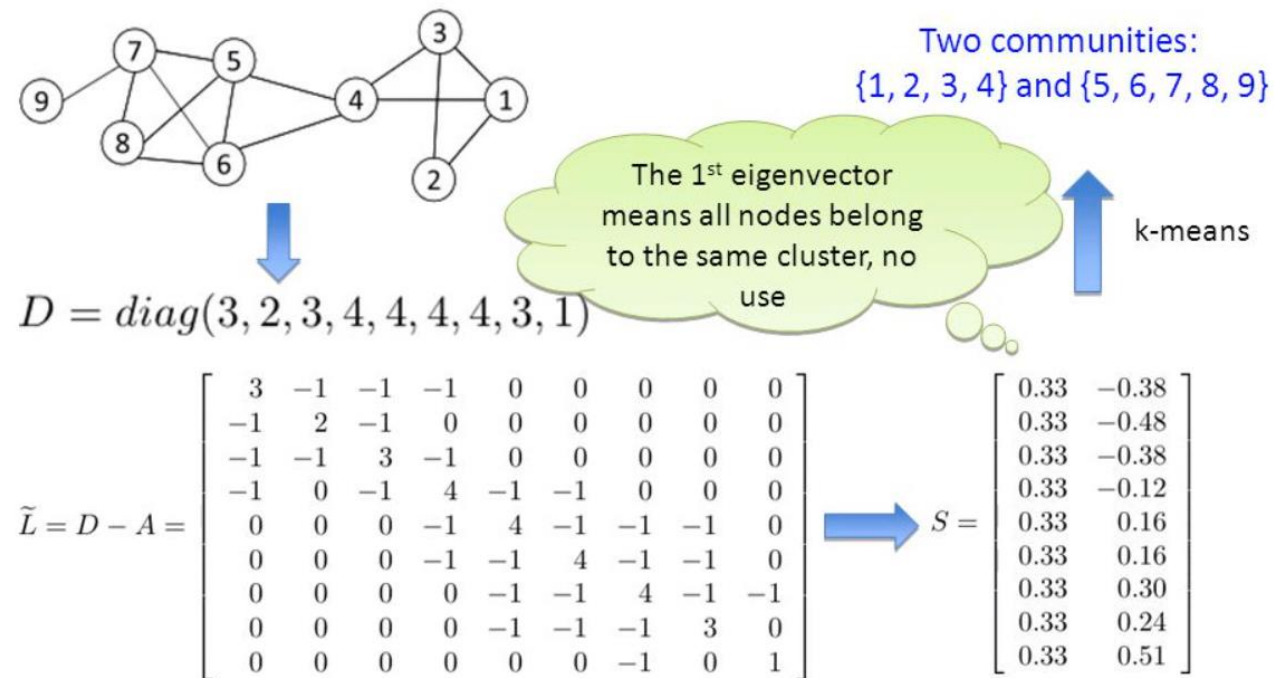
1. Create a similarity graph between our N objects to cluster.
2. Compute the first k eigenvectors of its Laplacian matrix to define a feature vector for each object.
3. Run k-means on these features to separate objects into k classes.



CLUSTERING AND COMMUNITIES ANALYSIS

■ Taxonomy community criteria

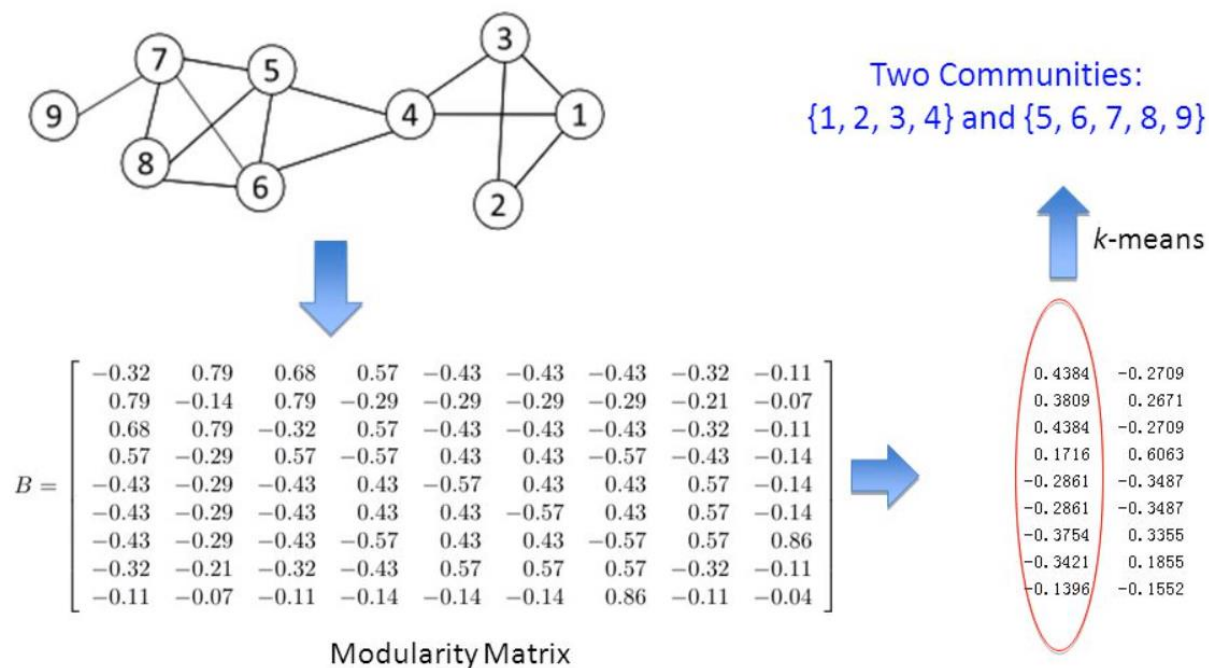
➤ Network-centric community detection: **Spectral clustering example**



CLUSTERING AND COMMUNITIES ANALYSIS

■ Taxonomy community criteria

➤ Network-centric community detection: **Modularity maximization example**

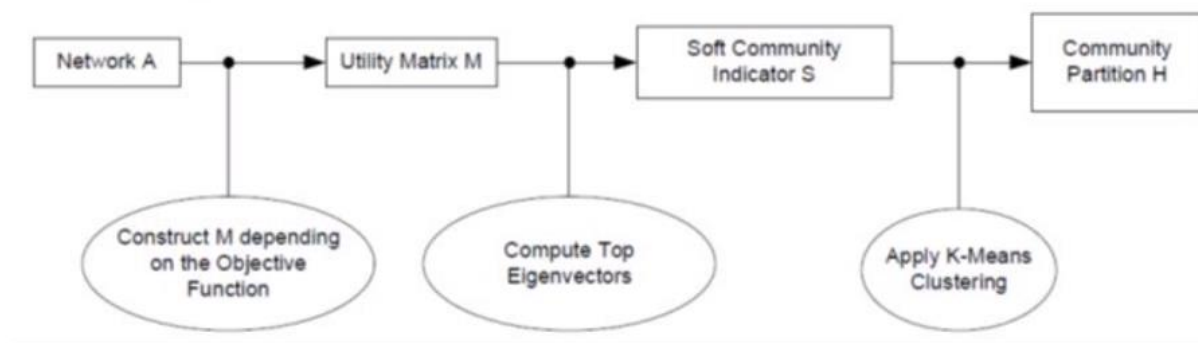


CLUSTERING AND COMMUNITIES ANALYSIS

■ Taxonomy community criteria

➤ Network-centric community detection: **A unified view for community partition**

- Latent space models, block models, spectral clustering, and modularity maximization can be unified as



$$\text{Utility Matrix } M = \begin{cases} \text{modified proximity matrix } \tilde{P} & \text{if latent space models} \\ \text{adjacency matrix } A & \text{if block models} \\ \text{graph Laplacian } \tilde{L} & \text{if spectral clustering} \\ \text{modularity maximization } B & \text{if modularity maximization} \end{cases}$$



CLUSTERING AND COMMUNITIES ANALYSIS

■ Taxonomy community criteria

➤ Hierarchy-centric community detection ([Source](#)):

- Goal: build a hierarchical structure of communities based on network topology
- Allow the analysis of a network at different resolutions
- Representative approaches:
 - Divisive Hierarchical Clustering (top-down)
 - Agglomerative Hierarchical clustering (bottom-up)



CLUSTERING AND COMMUNITIES ANALYSIS

■ Taxonomy community criteria

➤ Hierarchy-centric community detection: **Divisive hierarchical clustering**

◆ Divisive Clustering

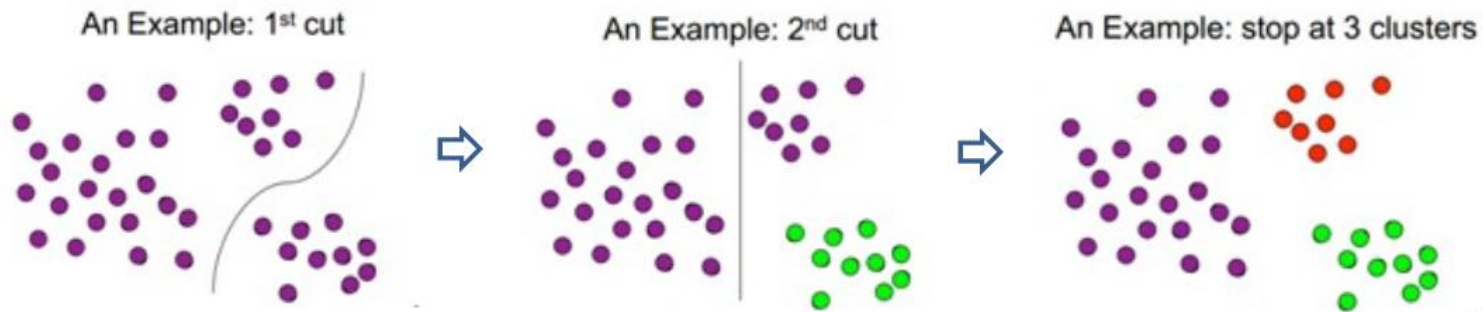
- 1. Put all objects in one cluster
- 2. Repeat until all clusters are singletons
 - a) choose a cluster to split
 - what **criterion**?
 - b) replace the chosen cluster with the sub-clusters
 - **split into how many**?



CLUSTERING AND COMMUNITIES ANALYSIS

■ Taxonomy community criteria

➤ Hierarchy-centric community detection: **Divisive hierarchical clustering**

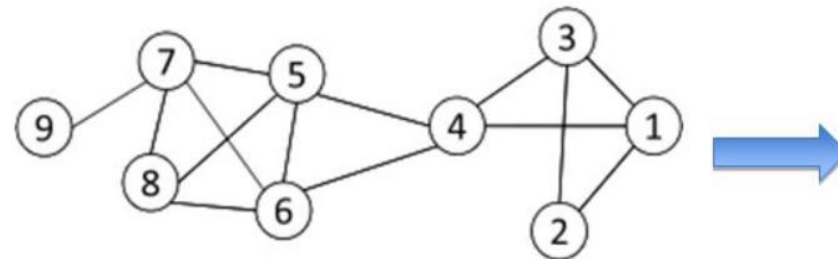


- A Method: Cut the “weakest” tie
 - At each iteration, find out the weakest edge.
 - This kind of edge is most likely to be a tie connecting two communities.
 - Remove the edge.
 - Once a network is decomposed into two connected components, each component is considered a community.
 - Update the strength of links.
 - This iterative process is applied to each community to find sub-communities.

CLUSTERING AND COMMUNITIES ANALYSIS

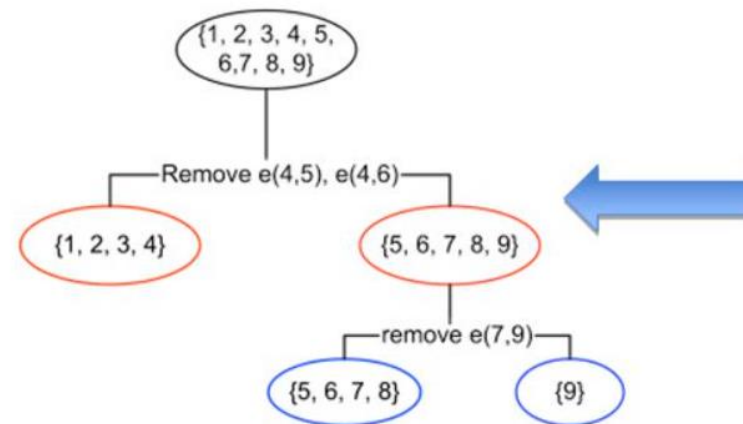
■ Taxonomy community criteria

➤ Hierarchy-centric community detection: **Divisive hierarchical clustering**



Initial betweenness value

	1	2	3	4	5	6	7	8	9
1	0	4	1	9	0	0	0	0	0
2	4	0	4	0	0	0	0	0	0
3	1	4	0	9	0	0	0	0	0
4	9	0	9	0	10	10	0	0	0
5	0	0	0	10	0	1	6	3	0
6	0	0	0	10	1	0	6	3	0
7	0	0	0	0	6	6	0	2	8
8	0	0	0	0	3	3	2	0	0
9	0	0	0	0	0	0	8	0	0



After remove $e(4,5)$, the betweenness of $e(4, 6)$ becomes 20, which is the highest;

After remove $e(4,6)$, the edge $e(7,9)$ has the highest betweenness value 4, and should be removed.

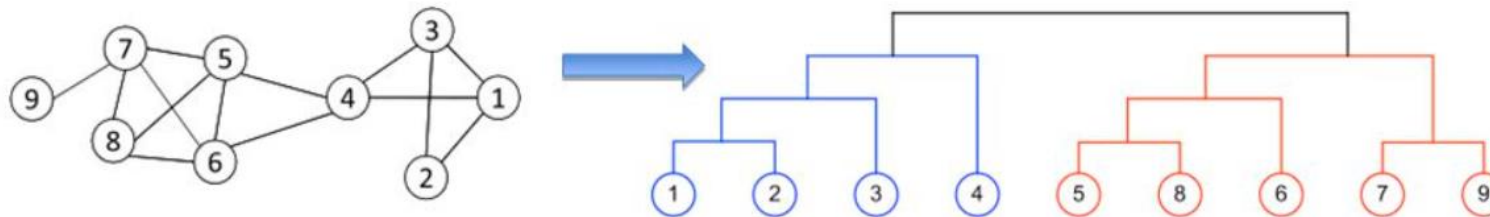


CLUSTERING AND COMMUNITIES ANALYSIS

■ Taxonomy community criteria

➤ Hierarchy-centric community detection: **Agglomerative hierarchical clustering**

- Initialize each node as a community
- Merge communities successively into larger communities following a certain criterion
 - E.g., based on modularity increase



Dendrogram according to Agglomerative Clustering based on Modularity

CLUSTERING AND COMMUNITIES ANALYSIS

■ Community evaluation

➤ Evaluating community detection

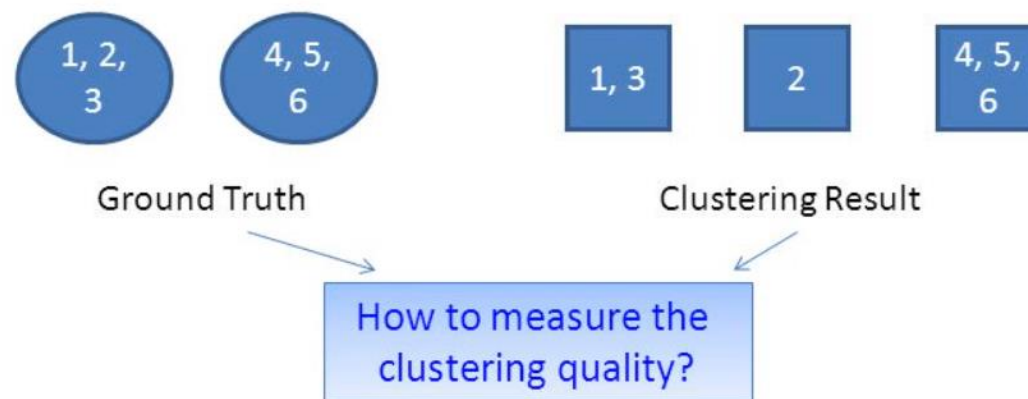
- For groups with clear definitions
 - E.g., Cliques, k-cliques, k-clubs, quasi-cliques
 - Verify whether extracted communities satisfy the definition
- For networks with ground truth information
 - Normalized mutual information
 - Accuracy of pairwise community memberships



CLUSTERING AND COMMUNITIES ANALYSIS

■ Community evaluation

➤ Measuring a clustering result



See more details
[here](#)

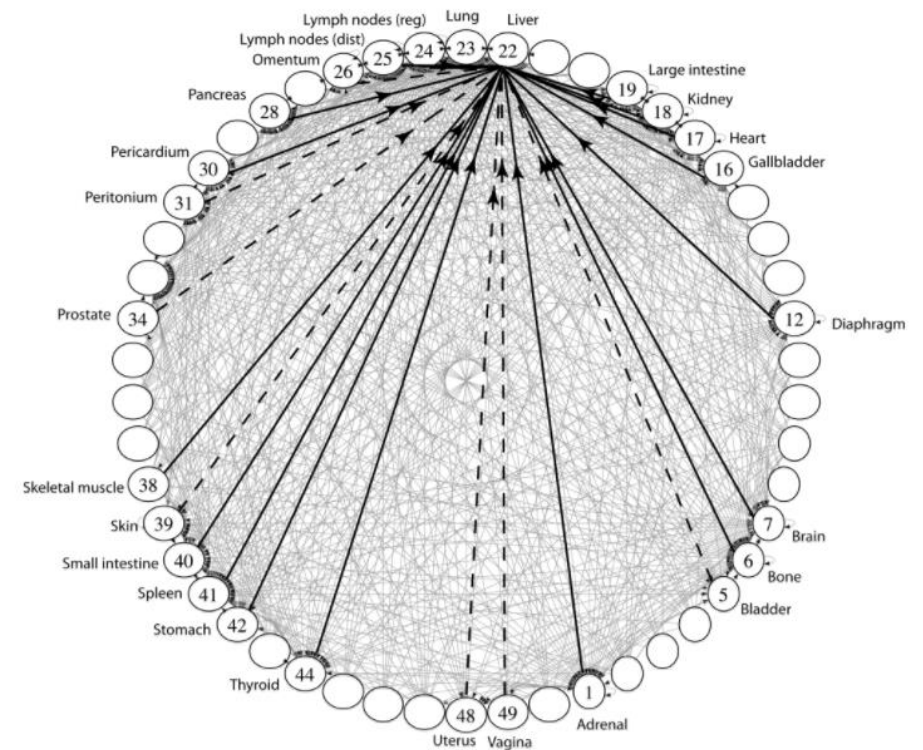
- The number of communities after grouping can be different from the ground truth
- No clear community correspondence between clustering result and the ground truth
- Normalized Mutual Information can be used

SOME APPLICATIONS

SOME APPLICATIONS

■ How researchers are fighting lung cancer using PageRank

- Main idea: Trying to predict the PageRank of tumors.
- The same types of tools that help us understand the spread of information through the web can help us understand the spread of cancer through the human body.
- E.g. a Kidney would likely have a higher PageRank than a liver because the kidney is more likely to spread cancer cells throughout the body



More information about the research application [here](#) and [there](#)



SOME APPLICATIONS

■ Social networks & e-Marketing

How to choose the blogger who can better speak of a product / service? Which is the best twitter user to follow?
To which customers is it interesting to propose a product in beta-test?

- Determine the individuals by whom the campaign should start / move:
 - Opinion leaders (popularity calculation - degree centrality, pageRank)
 - The mediators (calculation of "stress", "benneness" centrality): the actors who control the propagation of the information.
 - Reference points.
- Predict speed, broadcast coverage:
 - Calculation of shorter paths, maximum flows.
 - A rumor will be more effective if it comes from several different sources.



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